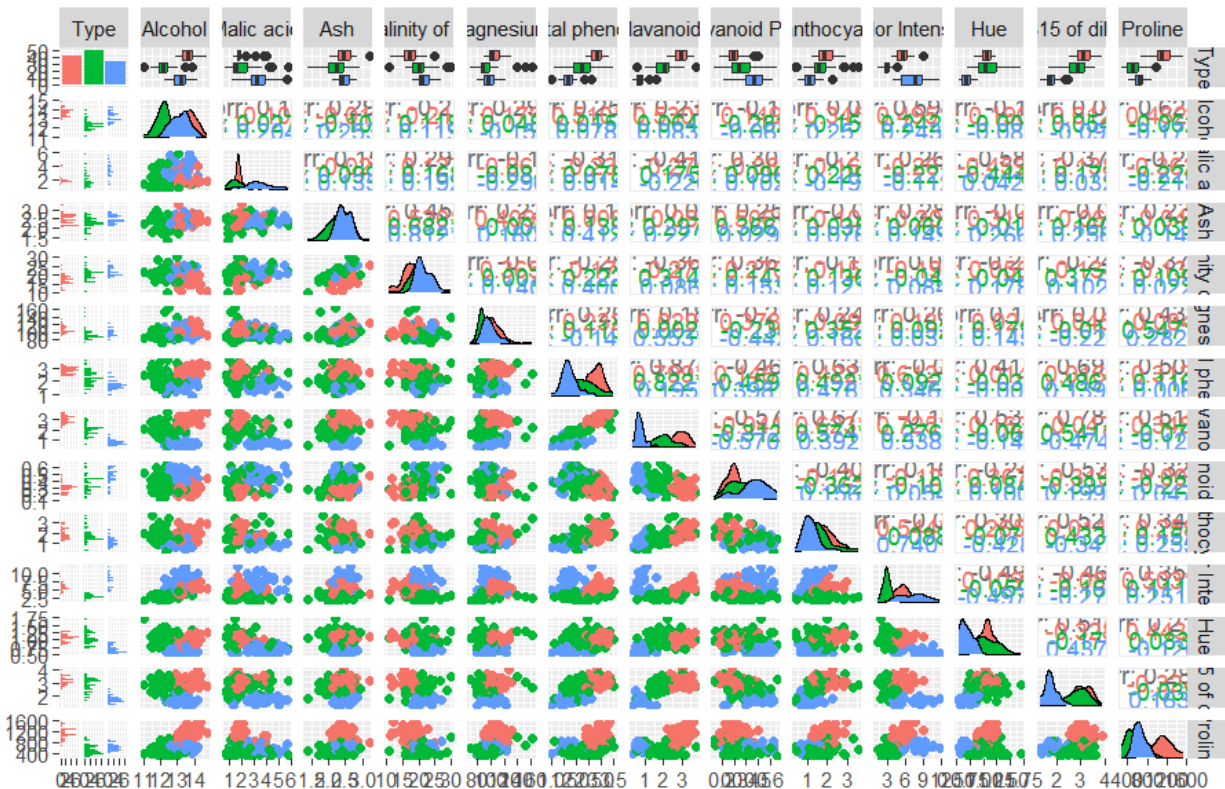


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Lab 5: Support Vector Machines

Results of a feature plot:



The summary of the first SVM model with a linear kernel:

```
> ## train SVM model - linear kernel
> svm.mod0 <- svm(Type ~ ., data = train, kernel = 'linear')
> svm.mod0
```

Call:

```
svm(formula = Type ~ ., data = train, kernel = "linear")
```

Parameters:

```
SVM-Type: C-classification
SVM-Kernel: linear
cost: 1
```

Number of Support Vectors: 20

The confusion matrix along with Precision/Recall/F1 scores of the SVM model with the linear kernel:

```
> cm = as.matrix(table(Actual = train$Type, Predicted = train.pred))
> cm
```

	Predicted			
Actual	1	2	3	
1	41	0	0	
2	0	50	0	
3	0	0	33	


```
> data.frame(precision, recall, f1)
```

	precision	recall	f1
1	1	1	1
2	1	1	1
3	1	1	1

The results of the tuned SVM with a polynomial kernel:

```
> tuned.svm
```

Parameter tuning of 'svm':

- sampling method: 10-fold cross validation

- best parameters:

gamma	cost
0.1	1

- best performance: 0.03205128

The results of the tuned SVM model with a polynomial kernel:

```
> svm.mod1
```

Call:

```
svm(formula = Type ~ ., data = train, kernel = "polynomial", gamma = 0.1,
     cost = 1)
```

Parameters:

SVM-Type:	C-classification
SVM-Kernel:	polynomial
cost:	1
degree:	3
coef.0:	0

Number of Support Vectors: 58

The confusion matrix and dataframe of the Precision/Recall/F1 scores of the tuned SVM model with a polynomial kernel:

```

> cm
      Predicted
Actual 1  2  3
      1 39  0  0
      2  0 47  0
      3  0  0 38
> data.frame(precision, recall, f1)
  precision recall  f1
1         1         1  1
2         1         1  1
3         1         1  1

```

Now, we will be comparing the confusion matrices and precision/recall/f1 dataframes for the testing set.

Model 1 (Linear Kernel):

```

> cm
      Predicted
Actual 1  2  3
      1 20  0  0
      2  0 23  1
      3  0  0 10
> data.frame(precision, recall, f1)
  precision  recall      f1
1 1.0000000 1.0000000 1.0000000
2 1.0000000 0.9583333 0.9787234
3 0.9090909 1.0000000 0.9523810

```

Model 2 (Tuned Polynomial Kernel):

```

> cm
      Predicted
Actual 1  2  3
      1 20  0  0
      2  0 24  0
      3  0  1  9
> data.frame(precision, recall, f1)
  precision recall      f1
1         1.00      1.0 1.0000000
2         0.96      1.0 0.9795918
3         1.00      0.9 0.9473684

```

Let's now compare with the random forest classifier:

```
> cm
```

	Predicted			
Actual	1	2	3	
1	20	0	0	
2	0	23	1	
3	0	0	10	

```
> data.frame(precision, recall, f1)
```

	precision	recall	f1
1	1.0000000	1.0000000	1.0000000
2	1.0000000	0.9583333	0.9787234
3	0.9090909	1.0000000	0.9523810

And a kNN classifier:

```
> cm
```

	Predicted			
Actual	1	2	3	
1	18	0	2	
2	0	15	9	
3	0	4	6	

```
> data.frame(precision, recall, f1)
```

	precision	recall	f1
1	1.0000000	0.900	0.9473684
2	0.7894737	0.625	0.6976744
3	0.3529412	0.600	0.4444444

Conclusion

From the above results, it is clear that the SVM models are more accurate in terms of classifying the type of wine based on all the features present in the dataset than random forest and kNN. However, the random forest model returned the same confusion matrix and precision/recall/f1 results as the SVM model with a linear kernel.